

Taresh Guleria

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PROFESSIONAL SUMMARY

Ph.D. Process Engineer, Industrial Engineering, Texas A&M (Dec 2026). Reduced qualification cycles ~80% and process waste ~75% via DOE and closed-loop control while maintaining >99% specification compliance. [O-1 & EB-2 NIW eligible]

EXPERIENCE

Graduate Research Assistant, Ph.D.

Jan 2023 – Present

Texas A&M University — Dept. of Industrial & Systems Engineering

College Station, TX

- Established production-scale defect-free process windows across platforms using DOE and statistical process control, driving 80% reduction in qualification coupon by eliminating rework through Lean/Six Sigma methodology.
- Engineered functionally graded structures with spatially-controlled transformation temperatures and elastic modulus gradients for precision sealing and damping applications – yielded first-pass qualification on Los Alamos containment project.
- Implemented in-situ thermal monitoring and closed-loop feedback framework in lab environment, improving build yield by ~60% through real-time defect detection and melt pool control; protocols now scaled to production builds.
- Documented process techniques and robustness protocols in ISO/ASTM AM standards contributions; established SOPs reducing field qualification time by 40% for new material submissions.

R&D Engineering Intern — Metals R&D & Government Projects

Jan 2026 – May 2026

EOS of North America

Pflugerville, TX

- Eliminated re-qualification bottleneck on DELTA-QUAL/ACCELERATE programs using in-situ process monitoring feedback and DOE parameters, further reducing 80% coupon testing with specs maintained & 65% throughput improvement.
- Head-started VITAL (LIFT program) for closed-loop process control and property-prediction framework, aligning qualification pathways and tool efficiency metrics across competing OEM platforms, further enabling cross-platform material certifications.
- Metal powder reuse and recycling methodology on RECAP (America Makes) program using LPBF laser parameter optimization and powder composition tracking, further cutting virgin feedstock consumption ~75% while maintaining tensile strength, fatigue life, and cyclic reliability specs under production loading conditions.

Research Intern (NSF INTERN Fellow) — Manufacturing Demonstration Facility

May 2024 – Aug 2024

Oak Ridge National Laboratory

Knoxville, TN

- Optimized LPBF parameters across 100+ samples to achieve >99% density and establish process–microstructure–property relations governing cyclic reliability under multiaxial loading.
- Deployed EBSD, in-situ XRD, and DIC full-field strain analysis to map defect-formation root causes and strain localization pathways, enabling predictive yield and fatigue-life models for production validation.

EDUCATION

Ph.D., Industrial Engineering, Texas A&M University, College Station, TX — GPA 4.0/4.0

Expected Dec 2026

M.S., Industrial and Systems Engineering, Texas A&M University, College Station, TX — GPA 3.45/4.0

Dec 2024

TECHNICAL SKILLS

Process Development & Manufacturing: Design of experiments (DOE), process window qualification, statistical process control (SPC), in-situ and in-line monitoring, closed-loop adaptive control, defect root-cause analysis, yield improvement, production readiness and process robustness validation

Metrology & Characterization: SEM, EBSD, in-situ XRD, DSC, DIC full-field strain, FEA thermomechanical, fatigue and cyclic testing, mechanical property prediction

Software & Tools: EOSPRINT, Dyndrite, 3DXpert, Siemens NX, ANSYS Additive Suite, CREO, AVEVA-PDMS, Python, MATLAB

SELECTED PUBLICATIONS & RECOGNITION

Publications: Google Scholar — Seven peer-reviewed articles in high-impact journals (*Additive Manufacturing*, *Acta Materialia*, *Materials Today*)

Recognition: 2nd Place of 80+ – ASTM AM Student Poster Competition, 2025 · NSF INTERN Fellowship – Oak Ridge National Lab, 2024 · Best Research Presentation – Additive Manufacturing Symposium, 2024.